

The time-one map Ψ is defined by setting $\Psi(x_0) = x(1)$ for each $x_0 \in \mathbb{R}^d$. Classical results on approximation properties of shallow neural networks do not immediately apply since $A(t)$ is fixed to be a square matrix of input dimension. For this reason, the neural ODE (1) is referred to as *narrow* following the definition in [2].

In [11], it has been shown that flows of this ODE can be used to approximate continuous maps on $\mathbb{R}^d$ in the uniform norm by considering a NODEs on $\mathbb{R}^{2d+1}$, provided the activation function satisfies a quadratic equation. In [9], it has been shown that any L_2 map can be approximated using a narrow NODE. The capability NODEs for transporting probability densities have been explored in [5, 9, 2]. For L_p maps, universal approximations have been shown in [4].

The strategy used in this paper departs from the techniques of the other cited works, where either the arguments are constructive or control theoretic arguments are used. Firstly, we note that it can be shown that flows of (1) can approximate flows of the wide NODE

$$\dot{x}(t) = \sum_{i=1}^m A_i \Sigma(W_i x + b_i) \quad x(0) = x_0 \quad (3)$$

This, in turn, can be used to show that flow maps of (1) can be used to approximate flows of any dynamical system of the form

$$\dot{x}(t) = V(x, t) \quad x(0) = x_0 \quad (4)$$

where $V(x, t)$ is a time dependent vector field and hence, $V(x, t)$ can be approximated by an arbitrarily wide network $\sum_{i=1}^m A_i \Sigma(W_i x + b_i)$ for any t . Therefore, narrow neural ODEs inherit the approximation properties of their shallow but wide counterparts. This strategy is used in [5] and in [6] to study flow approximation properties using narrow NODEs. In [6] it was in fact shown that the dimension of the weight parameters used to approximate maps can be taken to be $m \times d$, with m less than the input dimension d , by additionally exploiting geometric non-commutative properties of some chosen m basis vector fields, owing to diffeomorphism controllability results due to [1].

Our goal in this paper is to derive quantitative rates of approximation of (1). Existing quantitative rates are either established to approximate homoeomorphism in the L_2 norm [9] or for the purposes of generative modeling [9, 2]. In contrast, we are interested in quantitative rates for approximation of flow maps in the uniform norm. The problem we address is how many switches are needed in $A(t), W(t), b(t)$, to approximate the flow of (3). In this way, our result complements [9, 2]. While we restrict to reference flows of the differential equations of the form (3), an extension to general flows is straightforward since shallow but wide neural networks are known to be dense in the set of continuous functions. Hence, quantitative approximation properties of shallow neural networks would immediately translate to quantitative approximation properties of narrow NODEs.

Before we present our analysis, we give some brief intuition behind our proof strategy. The idea behind our analysis is the following. Given a differential equation of the